(a) maintain discipline whenever persons are riding in the shaft conveyance,
(b) enforce the specified load limits for the shaft conveyance, and
(c) notify the hoist operator if a heavy or an irregularly shaped load is being handled.

(2) Whenever persons are underground in a part of a mine served by a shaft hoist operated under manual control, the person in charge of the conveyance shall, when not riding in the conveyance, be available within sound of the shaft signals at all times.

Transport of Explosives

7.8.5 No person shall travel in a shaft conveyance that is carrying explosives, unless the person is required to handle the material and there is adequate space for the person's safety.

Notice Showing Conveyance Capacity

7.8.6 A notice clearly showing the load restrictions for the weight of material and the number of persons to be carried in the shaft conveyance shall be posted at the shaft collar and all entrances to the shaft.

Compartment Entry Prohibited

7.8.7 No person shall enter a hoisting compartment of a shaft in which hoisting operations are being carried on, except to enter or leave a shaft conveyance in that compartment.

Hoisting Suspended During Shaft Work

7.8.8 (1) Except during shaft sinking operations and concurrent preliminary development work, no hoisting operations shall be carried out simultaneously with any other work in a shaft compartment or in that part of the headframe used in conjunction with it.

(2) Where such work is to be done, the person in charge shall warn the hoist operator on duty and make a signed entry in the Hoist Operator's Logbook.

Shaft Examination

7.8.9 (1) No person shall work in a shaft or in that part of the headframe used in conjunction with it unless the person is adequately protected from accidental contact with a moving shaft conveyance and the danger of falling material or rock.

(2) No loading pocket shall be equipped with doors, gates, or chains which are closed by positive pressure, and which may open if the supply fails and discharge material into the shaft.

(3) No person shall work in a shaft compartment, or in a place immediately adjoining the compartment, or in that part of the headframe used in conjunction with it while hoisting operations are being carried on in the compartment, except where
(a) the shaft conveyance is necessary for carrying out the work, or
(b) the person is engaged in filling skips at a loading station and is properly protected in a location having adequate dimensions for safely carrying out the person's duties, or
(c) the work is being performed in a separate compartment that meets the requirement of section 7.1.5 (2).

Automated Hoisting

7.8.10 (1) Where a hoist is being operated by automatic or semi-automatic control and no other means are available for getting persons out of the mine quickly, a qualified hoist operator shall be readily available to operate the hoist whenever workers are underground.

(2) A device that permits changing from manual to automatic control shall be installed on an automatic hoist and the device shall be located where it is readily accessible to the manual controls, and operated only by an authorized person.

Automatic Hoisting – Location of Controls

7.8.11 (1) Where a hoist is designed to be operated automatically from controls located at shaft stations and inside the cage, the switch for effecting a changeover of the control made between the shaft station and the cage shall be effective only at the shaft station where the cage is stopped.

(3) Devices installed at shaft stations for selecting the cage destination and initiating hoist movement shall be effective only when the cage is stopped at that station.

(4) Except for jogging movements, a control located at a shaft station for initiating hoist movement shall be effective only when the shaft gate is closed.

(5) When an executive signal is given from a control at a shaft station, at least 5 seconds shall elapse before the hoist moves.

(6) Except for jogging movements, devices located inside a cage for initiating hoist movements shall be effective only when the doors of the cage and the shaft gates are closed.

(7) Where a control for initiating hoist movement is located inside a cage, a device shall be installed in the cage which can cause the hoist safety circuit to be interrupted in an emergency.

(8) If an emergency stop occurs during automatic or semiautomatic hoisting operations, an alarm shall sound and the hoist shall be operated manually until the reason for the emergency stop has been

found and, if caused by a fault, until the fault has been corrected and the hoist operated for sufficient time to ensure that it is operating normally.

Brake, Overwind and Underwind Testing

7.8.12 (1) After going on shift, and before raising or lowering a shaft conveyance, the hoist operator shall ensure that each drum brake is capable of holding its maximum permitted load by testing it against the normal, full load starting torque of the hoist. The drum shall not be unclutched until the test has been made.

(2) At least once in each 24-hour period of hoist operation, test the overwind and underwind protective devices by slowly raising or lowering the shaft conveyance or counterweight into them.

Hoisting Stoppages

7.8.13 A hoist operator shall make a return trip of the shaft conveyance

(1) through the working part of the shaft after any stoppage of hoisting for more than 2 hours, and

(2) below any part of a shaft that has been under repair, after the repairs have been completed.

Hoist Operator's Responsibilities

7.8.14 A hoist operator shall

(1) remain at the hoist controls at all times that the hoist is in motion under manual control,

(2) not talk with anyone while the hoist is in operation under the hoist operator's control, except in an emergency and during shaft maintenance and examination,

(3) not operate the hoist to transport any person unless at least two brakes can be applied to safely stop the conveyance,

(4) not lower persons on an unclutched drum, and

(5) whenever the hoist operator leaves the hoist controls, except when the hoist is put on automatic control, set the brakes and the controls so that at least two separate and distinct actions are required to put the hoist in motion.

Signals to Hoist Operator

7.8.15 (1) A hoist operator shall complete any hoist movement required by an executive signal without stopping unless the hoist operator receives a "stop" signal or an emergency occurs.

(2) After receiving a 3-bell signal, a hoist operator shall remain at the hoist controls unless advised orally by the person in charge of the conveyance that hoist movement will not be required.

Hoist Operator to Attend to Hoist Only

7.8.16 A hoist operator shall not be required to look after any other machinery except with the approval of the chief inspector.

Hoisting Prohibition

7.8.17 No person shall

(1) operate or interfere with any devices or controls for operating a hoist unless qualified and authorized,

(2) speak to the hoist operator while the hoist operator is operating the hoist on manual control, except in an emergency or when the hoist is being repaired, maintained, or adjusted and a notice to this effect shall be posted in the hoistroom,

(3) be on a cage while it is being placed onto or removed from chairs,

(4) be in, on, or under a shaft conveyance or counterweight which is supported by an unclutched drum unless the conveyance or counterweight is secured in position or unless permitted by section 7.1.22 (1), or

(5) leave a shaft conveyance that has inadvertently stopped at a point other than a shaft station, except upon instruction from a qualified and authorized person outside the conveyance.

Chairing Cage

7.8.18 Chairs shall not be used for landing a cage unless

(1) a signal for chairing has been given and returned, or

(2) special arrangements have been made to operate a cage with a car, in balance, from that location.

Object Falling Down Shaft

7.8.19 The normal operation of a hoist shall be suspended if an object, which may be a hazard to the operation of a shaft conveyance or counterweight, has fallen down the shaft until,

(1) a shaft inspection and a trial run through the affected part of the shaft have been made,

(2) any obstructions have been removed, and

(3) any damage affecting safe operation has been repaired.

Hoisting Procedures

7.8.20 (1) The manager shall issue written procedures for the safe operation of the hoist and have them posted in the hoist room.

(2) Every hoist operator shall be instructed in these procedures which shall include

(a) operating the hoist where there is an intermediate shaft obstruction,

(b) emergency procedures,

(c) commencing hoisting operations after an inadvertent stoppage of the hoist, and
(d) the operation of any safety devices.

Inadvertent Stoppage of Hoist

7.8.21 (1) The manager shall develop procedures for removing persons from a shaft conveyance that has inadvertently stopped at a place in the shaft other than a shaft station.

(2) The person designated to be in charge of a shaft conveyance by virtue of section 7.8.4, shall be readily available and trained to carry out these procedures.

Signal Required

7.8.22 No hoist operator shall move a shaft conveyance without receiving a proper signal unless the conveyance has been inadvertently stopped at a point in the shaft other than a station from which signals can be given, and the hoist operator has received instructions from a qualified and authorized person.

Record Keeping

Hoist Operator's Logbook

7.9.1 A hoist operator shall, each shift, enter in the Hoist Operator's Logbook

(1) the working condition of the hoist brakes, clutches, clutch-brake interlocks, depth indicators, hoist controls, signaling systems, overwind and underwind devices, and any other devices that may affect the safe operation of the hoist,

(2) any instructions the hoist operator received and any unusual circumstances that occurred affecting the normal operation of the hoist,

(3) the results of any tests or trial trips made in compliance with the code, and

(4) a report of any inadvertent stoppages that occurred.

Hoist Operator's Entries in Logbook

7.9.2 A hoist operator shall

(1) before commencing to operate the hoist, review and countersign all entries in the Hoist Operator's Logbook for the preceding two shifts,

(2) enter in the logbook the time at which the hoist operator commenced and finished the hoist operator's shift, and

(3) sign all entries made during the shift.

Hoist Operator's Logbook – Supervisor's Entries

7.9.3 (1) Any person issuing instructions to a hoist operator shall record and sign such instructions in the Hoist Operator's Logbook.

(2) Each working day, the supervisor in charge of a hoist shall review and countersign all entries made in the Hoist Operator's Logbook during the preceding 24-hour work period.

(3) The Hoist Operator's Logbook shall be kept in the hoistroom and be available for inspection at all times.

Rope Certificate

7.9.4 No shaft rope shall be used unless it is accompanied by a certificate from the manufacturer giving the following information

(a) name and address of the manufacturer, coil or reel number, date of manufacture and type, length, and diameter of rope,

(b) weight per unit of length, number of strands, number of wires in each strand, class of core, diameter of wires, breaking stress of steel from which the wire is made, and the results of torsion tests of individual wires, and

(c) its breaking load as determined by actual tests at a recognized testing laboratory.

Record Book

7.9.5 (1) The rope data required by section 7.9.4 along with the additional following information shall be entered in a book known as the Shaft Rope Record Book.

(2) The following information shall be included; date of purchase, date installed, identification number, name of shaft and compartment in which the rope is used, weight of shaft conveyance, maximum weight of material carried, weight of maximum length of rope in service, and static factor of safety.

History of Shaft Rope

7.9.6 (1) The Shaft Rope Record Book shall contain a history of each shaft rope including

(a) the date it was installed, shortened and removed, and reason for removal,

(b) dates and summaries of destructive and nondestructive tests, and

(c) certification of trial trips and examinations.

(2) The Shaft Rope Record Book shall always be available for inspection.

Non-destructive Tests: Hoisting Equipment

7.9.7 Without limiting section 7.4.6, every part of a hoist and hoisting equipment the failure of which could endanger persons, shall be subjected to an annual non-destructive test conducted by a certified person, and a copy of the test results shall be forwarded to an inspector.

Qualified Person to Be Appointed

7.9.8 The manager shall appoint a qualified person, or persons, to carry out the inspections prescribed in sections 7.9.9, 7.9.10, 7.9.11, 7.9.12 and 7.9.13 of the code.

Daily Inspection

7.9.9 At least once in each normal production day, an inspection shall be carried out of

(1) the exterior of each hoisting rope and tail rope to detect the presence of kinks or other visible defects and to note the condition of the rope dressing, and

(2) the cage safety catches for any visible damage or defects.

Weekly Inspection

7.9.10 At least once in each week, an inspection shall be carried out of

(1) all shaft conveyance safety mechanisms for proper adjustment, freedom of movement, and freedom from damage,

(2) all head, deflection and idler sheaves, and shafts and their bearer and sole plates,

(3) all rope attachments, shaft conveyance and counterweight attachments, and suspension gear,

(4) any shaft conveyance counterweight and work platform,

(5) all parts of the hoist that could affect its normal operation including the brakes, clutches, interlocks, depth indicators, and all safety devices,

(6) any hoisting equipment being used during shaft sinking operations, and

(7) any auxiliary brake operating weights to ensure that they move freely and have adequate holding capacity.

Monthly Inspection

7.9.11 At least once each month, an inspection shall be carried out of

(1) the shaft ropes to determine the amount of wear, distortion and corrosion, the need for lubrication, and the need for changing any wear patterns,

(2) the hoisting ropes to determine the number of broken wires, and

(3) the friction treads on a friction hoist.

Six-monthly Inspection

7.9.12 At least once in every 6 months of service, an inspection shall be carried out of

(1) the hoisting rope on a drum hoist within the attachments at the drum and at the drum spout, and

(2) the hoisting rope of a friction hoist within the attachments at the shaft conveyance and counterweight in accordance with a procedure established by the manager.

Annual Inspection

7.9.13 At least once in every 12 months, an inspection shall be carried out of,

(1) foundation bolts, bolt locking devices, and all other bolts and fastenings that are critical for hoist safety, and

(2) the bails, suspension gear and structural components of every shaft conveyance and counterweight.

Testing Safety Catches

7.9.14 (1) A qualified and authorized person shall, at least once in every month, test the safety catches on a shaft conveyance by suddenly releasing the empty conveyance from rest so that the safety dogs have the opportunity to grip the shaft guides and hold the conveyance.

(2) At least once in each year, the safety catches shall be subjected to a quick release test with the conveyance carrying a load equivalent to the maximum allowable load for that conveyance.

(3) If the safety catches do not operate satisfactorily, the shaft conveyance shall not be used for the transport of persons until the necessary repairs have been made and the safety catches re-tested and found to be satisfactory.

Hoist Rope Maintenance

7.9.15 (1) Hoisting ropes in use on a drum hoist shall be cleaned when necessary and be dressed with lubricant at least once each month so as to maintain a good coating.

(2) The supervisor in charge of the work shall enter the date and sign a record of the rope cleaning and dressing in the Hoisting Machinery Record Book.

Special Requirements for Ends of Rope

7.9.16 After every eighteen months of service on a friction hoist, the portion of a hoisting rope or tail rope that is within a wedge and socket attachment shall be cut off unless that portion of the rope is visually examined by a qualified and authorized person and found not to have

(1) any broken wires, significant corrosion, or serious pitting, and

(2) any excessive deformation of the wires.

Hoisting Machinery Record Book

7.9.17 A record of all inspections and examinations carried out in compliance with Part 7 of the code on the hoisting equipment, and a record of all repairs and maintenance work done, shall be entered in the Hoisting Machinery Record Book and all entries shall be dated and signed by the person who performed the work.

Record of Mechanical Failures and Accidents

7.9.18 (1) A record of any failure or accident involving a mechanical part of a hoist installation shall be made in the Hoisting Machinery Record Book by the supervisor responsible for the mechanical parts of the hoisting plant.

(2) The supervisor responsible for the mechanical parts of the hoisting plant shall countersign each entry made in the Hoisting Machinery Record Book with respect to all examinations and inspections carried out in compliance with Part 7 of the code.

Electrical Inspections

7.9.19 (1) A qualified and authorized person or persons shall be appointed by the manager to examine, inspect, and test the electrical parts of an electrically powered or controlled hoist.

(2) The person or persons appointed shall, at least once each week, make an examination of the hoist motors, the hoist controls, the electrical safety devices, and the signaling system.

(3) A record of the examinations and of all repair and maintenance work carried out shall be made in the Electrical Hoisting Equipment Record Book.

(4) All entries in the Electrical Hoisting Equipment Record Book shall be dated and signed by the person performing the work.

Record of Electrical Failures and Accidents

7.9.20 A record of any failure or accident involving an electrical component of a hoist motor, a hoist control, an electric safety device, or the signaling system shall be made in the Electrical Hoisting Equipment Record Book by the supervisor in charge of the electrical parts of the hoisting equipment.

Hoisting Plant Supervisor's Responsibilities

7.9.21 The supervisor in charge of all of the hoisting plant and equipment shall

(1) at least once each week, review the entries made in the Hoisting Machinery Record Book and the Electrical Hoisting Equipment Record Book during the preceding week,

(2) ensure that all examinations and tests, required by Part 7 of the code, have been carried out and any necessary repairs or adjustments made, and

(3) upon completion of the supervisor's review, certify in each record book that the supervisor has complied with subsections (1) and (2).

Shaft Inspection

7.9.22 A qualified and authorized person shall carry out an examination

(1) at least once every week, of each mine shaft,

(2) at least once every month, of the shaft guides, timbers, walls, and compartments used for hoisting,

(3) at least once every year, of the headframe, headframe foundations and back legs, sheave decks, dumps, bins, and bin supports,

(4) of the shaft sump at such frequency as is necessary to ensure that the tail, guide, and rubbing rope connections are clear of water and spillage, and

(5) at least once every year, of the water in the shaft sump to determine the pH value.

Record of Shaft Inspection

7.9.23 A record of the examinations required by section 7.9.22, and a record of all maintenance and repair work performed shall be entered in the Shaft Inspection Record Book and all entries shall be dated and signed by the person who carried out the work.

Supervisor of Shaft and Headframe Responsibilities

7.9.24 The supervisor in charge of the mine shaft and headframe shall

(1) at least once each week, review the entries made in the Shaft Inspection Record Book during the preceding week,

(2) ensure that all examinations and tests required by Part 7 of the code have been carried out and any necessary repairs or adjustments made, and

(3) upon completion of the supervisor's review, certify in the Shaft Inspection Record Book that the supervisor has complied with subsections (1) and (2).

Appendix 7-1

Mine Shaft Signal Code

1 bell STOP IMMEDIATELY – if in Motion

1 bell HOIST

2 bell LOWER

3 bell PERSONS ABOUT TO ENTER OR LEAVE CONVEYANCE

(1) The 3-bell signal shall be given before persons are permitted to enter or leave the shaft conveyance.

(2) Where a return bell signal system is installed, the hoist operator shall return the 3-bell signal before persons are permitted to enter or leave the shaft conveyance.

(3) A hoist operator who has received a 3-bell signal shall remain at the hoist controls until receipt of the signal designating the movement required and completion of the movement.

(4) The hoist operator shall initiate a 3-bell signal as the shaft conveyance approaches the level.

4 bell BLASTING SIGNAL

(1) The hoist operator shall answer a 4-bell signal by raising the shaft conveyance a few feet and then lowering it slowly.

(2) Following a 4-bell signal, only a one bell signal shall be required to signal for raising workers away from a blast.

(3) The hoist operator shall remain at the controls until the act of raising has been completed.

5 bell RELEASE SIGNAL

The hoist operator on receiving a 5-bell signal may move the shaft conveyance to another point in the shaft, not a recognized stopping point and stop it there at the hoist operator's own discretion, but the person giving the release signal shall remain to guard the conveyance until it is moved.

9 bell DANGER SIGNAL

This signal shall be given only in case of fire or other danger, and followed by the signal for the level at which the fire or other danger exists.

1 bell followed by 2 bell – CHAIRING

3 bell followed by 3 bell followed by 1 bell – HOIST SLOWLY

3 bell followed by 3 bell followed by 2 bell – LOWER SLOWLY.

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Magazines

Permit Required

8.1.1 The manager shall

(1) apply for and receive an Explosives Storage and Use Permit from the inspector before a magazine is located, erected, built, put into service, or modified, or before carrying out any maintenance work including the installation of lighting or heating, and

(2) ensure that the magazine meets the regulations and standards of the *Explosives Act* (Canada).

Magazine Location

8.1.2 (1) The manager shall select the site for a magazine in accordance with *Explosives Act* (Canada) and NRC Explosives Branch.

(2) The manager shall cease to use the magazine for the storage of explosives if the conditions under which the permit was issued are no longer valid.

(3) Upon taking a magazine out of service the manager shall

(a) return to the inspector, by registered mail, the storage and use permit for the magazine taken out of service,

(b) remove all explosives from the magazine, and

(c) dispose of all unused explosives in a suitable manner.

Electrical Specifications

8.1.3 In every magazine the electrical equipment and wiring shall meet the requirements of CSA Standard M421-00 and NRCan "Storage Standards for Industrial Explosives.

Flammable Material

8.1.4 Gasoline, oil, or other flammable material shall not be stored in a magazine, or nearer to a magazine than the distance required by the Table of Distances in NRC Explosives Branch "Blasting Explosives and Detonators Storage, Possession, Transportation, Destruction and Sale," or as directed by the chief inspector.

Magazine Signs

8.1.5 All magazines must have suitable signs

(a) located in a manner that does not attract undue attention, and

(b) prohibiting open flames, or any ignition source including smoking within 10 metres.

Magazine Rules

8.1.6 Every magazine shall be operated and maintained in accordance with the NRC regulation and the following rules

(1) it shall be in the charge of a authorized person who shall carry out a weekly inspection of the magazine,

(2) it shall be locked at all times, except when explosives are being moved, and only the authorized person(s) are in possession of the key,

(3) it shall have an up to date inventory of its contents in a special logbook and all entries shall be signed by the authorized person in charge,

(4) it shall be kept clean, dry, and free from grit at all times,

(5) it shall be kept free of broken explosives packages or spilled explosives, and when necessary, the shelves and floors shall be treated with a suitable neutralizing agent to remove all traces of explosive substances,

(6) it shall have its contents arranged in a tidy and organized manner, including any explosives returned from a workplace,

(7) it shall not contain any exposed iron or steel except in fixtures, and

(8) stock shall be rotated, so for each type and size of explosive, the oldest stock is used first.

Posting Rules

8.1.7 A copy of the rules for magazine operation and maintenance, required by section 8.1.6, shall be posted inside every magazine.

Storage of Detonators

8.1.8 (1) Detonators shall be stored in a special, separate building designated as a "Detonator House", to which the rules for magazines required by section 8.1.6 shall apply.

(2) Detonators shall not be kept or stored with explosives or taken into any magazine containing explosives.

(3) In an underground mine with the permission of the inspector, detonators may be kept or stored in suitable, closed containers which shall not be placed closer than 10 m to any explosives except during transport or when priming.

No Smoking

8.1.9 No person shall smoke while handling, transporting or using explosives.

Underground Explosive Storage

8.1.10 In an underground mine, other than a coal mine

(1) A magazine for the storage of more than a 24 hour supply of explosives may be established with the written approval of the inspector.

(2) Explosive storage boxes, holding not more than a 24 hour supply, may be maintained in each working place without the permission of an inspector. The boxes shall be clearly marked "Explosives."
   (a) A current record of every magazine or explosive storage area shall be kept.

(3) A magazine, explosive container, or explosive storage area shall be
   (a) located at least 100 metres from a shaft, hoist room, main access ramp, refuge station, transformer vault or a fuel storage or fuel transfer area, and
   (b) located so that there is no possibility of a vehicle colliding with a storage container, and
   (c) located so that in case of fire in the mine the explosives are not likely to become overheated, and
   (d) conspicuously marked by "Explosives" sign or signs.

(4) Detonators are not to be stored within 10 metres of explosives, and

(5) Where longhole blasts or similar blasting operations are being carried out, such quantities of explosives as can be loaded in a 24 hour period together with an amount that may be necessary to maintain that supply may be kept in a designated storage area that is not a magazine which is under the control of an authorized person.

Blasting Certificates

Blasting Certificate Required

8.2.1 (1) No person shall conduct, or be permitted to conduct, a blasting operation unless the person holds a provisional or valid blasting certificate granted under this part of the code.

(2) No person shall be granted a provisional or valid blasting certificate unless the person is able to give and receive orders in the English language.

Assistance for Blaster

8.2.2 A person who does not hold a blasting certificate shall not prepare a blast, but a certificated blaster may be assisted by a reliable person not holding a blasting certificate. The blaster shall ensure that the person assisting the blaster remains under the blaster's supervision.

Provisional Blasting Certificates

8.2.3 A provisional blasting certificate, valid for a period not exceeding 90 days, may be granted by the manager, on a one time basis, and a copy of the certificate shall be sent to the inspector.

Blasting Certificate Examination

8.2.4 An applicant for a blasting certificate shall pass a written examination, and satisfy the inspector that the applicant is qualified to perform the duties of a blaster.

Certificate Restrictions

8.2.5 (1) A blasting certificate may contain any restriction deemed necessary by the inspector.

(2) The blaster shall deliver the blaster's certificate to the manager when the blaster commences employment at the mine and the manager shall return it to the blaster when the blaster's employment is terminated, unless the certificate has been suspended under section 8.2.6 or 8.2.7.

(3) The blasting certificate shall be valid for a period of 5 years.

Blasting Certificate Suspension by Manager

8.2.6 The manager shall suspend a person from working with explosives if, in the manager's opinion, the person fails to comply with any section of the code related to blasting, commits a careless act with explosives or detonators, or is unfit to perform the person's duties as a blaster, and the manager shall notify the inspector of the suspension without delay.

Blasting Certificate Suspension by Inspector

8.2.7 An inspector may suspend or cancel a blasting certificate if, in the inspector's opinion, the holder fails to comply with any section of the code related to blasting, commits a careless act with explosives or detonators, or is unfit to perform the holder's duties.

Use, Care and Handling of Explosives

Acceptable Explosives

8.3.1 Explosives used in a mine shall be licensed, as per the *Explosives Act* (Canada), and they shall be used in accordance with the manufacturers recommendations.

Identification of Explosives

8.3.2 All explosives shall have plainly marked or printed on the exterior of every original package.

(1) the name and place of business of the manufacturer and the date of its manufacture,

(2) the type and strength of the explosive, and the dimensions of the cartridge,

(3) the UN classification number and the dangerous goods classification, and

(4) when full cases of detonators, detonating cords, and boosters are received, the inner cartons are immediately marked with the magazine license number.

Opening Containers

8.3.3 Only implements made of non-sparking material shall be used to open cases containing explosives.

Defective Explosives

8.3.4 Any explosives believed to be defective shall

(1) if in a workplace, be immediately removed,

(2) be reported to the manager and the inspector, and

(3) be destroyed in a safe manner in accordance with the manufacturer's recommendations.

Safety Fuse Assemblies

8.3.5 The manager must provide a written procedure for the use of safety fuse.

Frozen Explosives

8.3.6 Frozen explosives shall not be loaded until the product has been thawed in a manner recommended by the manufacturer. They shall not be warmed near an open fire or a steam boiler, nor by direct contact with steam or hot water.

Blasting Heated Rock

8.3.7 Where the temperature of the rock or material to be blasted exceeds 65 degrees Celsius, the manager shall ensure that special procedures and precautions are developed and these procedures shall be forwarded to the OHSC.

Transporting Explosives

8.3.8 Explosives and detonators

(1) shall be carried or transported in the manner prescribed in this code, and

(2) if not used, shall be returned to the magazine or storage box, and shall not be hidden in or about the mine.

Careless Acts

8.3.9 (1) No person shall

(a) commit a careless act with explosives or detonators, or

(b) having discovered that such an act has been committed, neglect or omit to report immediately the incident to the shiftboss, who shall report the matter forthwith to the manager.

(2) The manager shall report the offence, without delay, to the inspector and to the OHSC.

Blasters Medical Surveillance

8.3.10 Blasters that are continually handling nitro-glycerine based explosives shall have a medical examination on a frequency of less than 3 year intervals to determine sensitivity to and any harmful health effects from the continued exposure to nitroglycerine.

8.3.11 Electric detonators shall not be used when extraneous currents (stray currents) exceed 50 milliamps (.05 amps).

Transportation of Explosives

Supervision by a Competent Person

8.4.1 (1) Only a person authorized by the manager, shall remove explosives from a magazine and transport them to a workplace, and such a transfer shall be made without undue delay.

(2) No person shall smoke or have open-flame lights within 10 m of a vehicle transporting explosives.

Vehicle Requirements

8.4.2 A vehicle used to transport explosives shall

(1) have a separate compartment for the explosives which prevents them from coming into contact with any metal that could produce a spark,

(2) be constructed so that the explosives cannot fall from the vehicle,

(3) when carrying explosives, be provided with orange diamond-shaped placards and clearly visible signs marked “Explosives” in letters not less than 150 mm in height which are posted on the front, rear, and sides of the vehicle,

(4) not be refueled when carrying explosives, except in an emergency,

(5) have its engine shut off while loading or unloading explosives except where the vehicle uses an engine-powered device for loading and unloading the explosives,

(6) only be operated by an authorized person, and

(7) be equipped with suitable fire extinguishing equipment.

Transport of Detonator

8.4.3 A vehicle used to transport explosive material at a mine shall only carry detonators when the detonators are separated from other explosives by a solid partition of wood 15 centimetres thick and extending at least 15 centimetres above the highest level to which explosives are packed in the vehicle.

Transport by Locomotive

8.4.4 When explosives or detonators are being transported by means of a locomotive they shall

(1) not be carried on the locomotive,

(2) be protected from any trolley wires, batteries or other hazards, and

(3) not be transported in front of the locomotive unless a worker walks in front to effectively guard them.

8.4.5 The person in charge of moving explosives or detonators to a headframe for transporting underground shall, immediately before doing so, notify the hoist operator, the deck person, and the cage tender.

Separate Containers

8.4.6 Explosives and detonators shall be transported in separate containers.

Loading and Priming

Care in Loading

8.5.1 (1) Only loading tools made of non-sparking materials shall be used.

(2) Explosive shall not be loaded into a hole

(a) of insufficient size, or

(b) by hitting, pounding, ramming, or applying undue pressure.

Cartridge Wrapper

8.5.2 Cartridge explosives shall not be removed from their wrappers except for water-gel or emulsion explosives that are to be used for blasting oversize rock or bringing down hung-up material.

Priming Nitroglycerine Explosives

8.5.3 Primers shall be made up only as required and, when priming with nitroglycerine type explosives, only a non-sparking implement shall be used to punch a hole in the explosive.

Priming Detonating Devices

8.5.4 Every primed charge shall

(1) contain a properly prepared detonating device sufficient to initiate the explosion reliably,

(2) have the detonator inserted into the cartridge in such a manner that it cannot be separated or pulled out of the cartridge during the loading operation, and

(3) be fired electrically or by approved non-electric means acceptable to the inspector.

Carrying Capped Fuses

8.5.5 A person may carry capped fuses with explosives from the nearest storage place to a point of use, without placing them in a container, provided that they are kept separate from each other. In no case, however, shall primed cartridges be transported.

Pneumatic Loading

8.5.6 Pneumatic loading of explosives into blastholes shall only be carried out

(1) with explosives licensed for that purpose, as per the *Explosives Act* (Canada), and

(2) where the procedures and equipment used will prevent any dangerous build-up of static electricity or hazards from stray electric currents.

8.5.7 The written approval of an inspector shall be obtained before any pneumatic loading of explosives is carried out over the leg wires of detonators.

Extraction of Explosives

8.5.8 No person shall extract, or attempt to extract, any primer or explosives of the nitroglycerine type from a loaded blasthole.

(1) Explosives of the ammonium nitrate type may only be removed from a blasthole by washing.

(2) Insensitive water-gel or emulsion type explosives may be blown out of a blasthole by an authorized person using moderate air pressure and a blowpipe made of non-sparking material.

Vehicles Prohibited

8.5.9 A blast area shall be clearly identified by posted signs to prevent the inadvertent access of vehicles.

8.5.10 No equipment shall be allowed within 8 m of any charged blasthole except

(1) authorized explosives vehicles,

(2) explosive vehicles which have the exhaust directed above the cab of the vehicle, and

(3) other equipment authorized in writing by the chief inspector.

Firing Explosives

Time for Blasting

8.6.1 The time for blasting shall be set so as to protect persons from exposure to dust, fumes, and smoke.

Guarding

8.6.2 Before blasting, the blaster, or person designated responsible for blasting, shall

(1) clear the blast danger zone of all persons,

(2) ensure that all entrances to the blast danger zone are guarded to prevent access, and not rely only on signs for the purpose, and

(3) when blasting on the surface of a mine, ensure that an effective audible warning is given.

Firing Procedure

8.6.3 (1) Every charged hole shall be fired in its proper sequence, and where any blast could affect other charged blastholes, all of the holes shall be fired in one operation.

(2) When required, a waterspray shall be used during blasting of development headings.

Waiting Time

8.6.4 (1) A blaster shall allow sufficient time between blasts in an underground mine for

(a) the heading, or other workplace, to be cleared of fumes, and

(b) the broken rock to cool sufficiently so that there is no danger when it is being loaded.

(2) A worker shall not return, or be permitted to return, to a workplace after a blasting operation until the worker is sure that the gases produced by the explosives have been removed or diluted to a "safe degree" by the ventilation system. If the worker has reasonable grounds to believe that this has not been achieved, the worker shall request the shiftboss to test the air with a suitable instrument and explain the results to the worker.

Reporting Misfires

8.6.5 A blaster shall

(1) when possible, count the number of shots exploding,

(2) report to the shiftboss if the blaster believes that any shot did not fire, and

(3) identify any misfired hole by inserting a conspicuous, non-metal marker at its outer end, or by roping the area off or by any other manner approved by the shiftboss.

Misfires

8.6.6 In the event of a misfired shot, the shiftboss shall

(1) determine the location, direction and depth of any hole necessary for blasting the misfired shot, supervise its drilling, and

(2) record in the daily examination and report book, the location of any misfired shot remaining at the end of the shift.

Raising and Sinking

8.6.7 Blasts shall be fired from a remote location during sinking operations, and when raising has exceeded 10 m or when required by the inspector.

Procedures for Explosives and Accessories

8.6.8 Written procedures shall be established for the use of all explosives accessories and blasting machines following the manufacturer's recommended specifications.

Blasting Machines

8.6.9 (1) All blasting machines shall be suitable for their applications, kept in a cool, dry, storage place and maintained in serviceable condition, in accordance with the manufacturers recommendations.

(2) The power output of every electric blasting machine shall be tested by an authorized person, in the manner prescribed by the manufacturer, at least once each month, and immediately before being returned to use after having been taken out of service.

(3) The results of the tests carried out in compliance with subsection (2), shall be entered in a logbook, dated, and signed by the authorized person who made the tests.

Condenser Discharge Machines

8.6.10 Condenser discharge machines shall be examined to ensure that any residual charge remaining on the capacitors after use is discharged in accordance with the manufacturer's instructions.

Connection to the Blast Circuit

8.6.11 (1) A blasting machine shall not be connected to the blasting cables until immediately before firing the charges.

(2) A blasting machine shall be immediately disconnected from the blasting cables after firing or attempting to fire the charges.

Blasting From Power Lines

8.6.12 (1) Electric power from lighting or power circuits shall not be used for firing charges unless

(a) the blasting circuit has an isolating transformer, and

(b) a special firing device that opens the blasting circuit by gravity is used.

(2) The blasting circuit conductors between the firing device and the blast site shall be No. 12 AWG, or better, and readily identifiable as blasting cable.

(3) Where expendable connecting wire is used, it shall not be less than No. 20 AWG.

Blasting Switches

8.6.13 Every electric power line blasting switch shall

(1) have the live side of the device installed in a box which is fixed, locked, and accessible only to the blaster, and

(2) incorporate a lightning gap of at least 1.5 m between the blasting switch and the service switch. This gap shall only be closed by a twist-type plug and cord assembly immediately before firing.

Branch Circuits

8.6.14 Where a blasting circuit is used for more than one working place, each branch circuit shall be isolated by means of a locked isolating switch that automatically short circuits the branch circuit.

Circuit Testing

8.6.15 Where more than one shot is fired electrically, the blaster shall test the electrical circuit with an approved circuit testing device immediately before blasting.

Electric Storms

8.6.16 The manager shall ensure that persons engaged in blasting operations underground are warned of an electric storm.

8.6.17 A blaster shall not connect an electric blasting circuit during an electrical storm. Once an electric blasting circuit is connected, and should an electric storm occur, all persons shall be withdrawn to a safe distance from the blast site. Access to the blast site shall be guarded until the storm has passed.

Radio Transmitters

8.6.18 The blaster shall ensure that signs are posted to ensure mobile radio transmitters shall be turned off when within 20 m of a site where an electric blasting system has been connected.

Blasting Cables Near Electrical Conductors

8.6.19 Blasting circuit conductors shall be kept at least 150 mm away from power and lighting cables and from any other electrical conductors.

Blasting Lead Wires

8.6.20 All blasting circuit conductors leading to a blasting site shall be insulated and, except when firing the blast, shall be short circuited.

Return to the Work Area

8.6.21 When blasting by electricity, the blaster shall not enter, or allow any other person to enter the area until

(1) the firing cables have been disconnected from the firing device and short circuited, or

(2) in the case of a blasting operation using a power or lighting circuit, the switches of the blasting circuit have been locked in the open position.

Firing in Multiple Work Places

8.6.22 Blasting cables or wires that have been used for blasting in one workplace, shall not be used for blasting in another workplace until all proper precautions have been taken to ensure that the blasting cables or wires have no electrical connection with the leads from the first workplace.

Central Blasting

8.6.23 A written summary of the procedure to be followed and a layout of the system shall be submitted to an inspector for approval before a central blasting system is used or modified. The procedure shall include a provision for post blast inspection.

Drilling Near Explosives

Misfired Holes and Bootlegs – Drilling Precautions

8.7.1 Drilling shall not be carried out

(1) within 300 mm of a bootleg on the surface or within 150 mm underground, or

(2) where any part of the hole to be drilled could come within 5 m of a hole containing explosives unless it is drilled under the direct supervision of a shiftboss to clear a blocked hole, or to place an additional hole for blasting a misfired charge in accordance with section 8.6.6 (1).

8.7.2 (1) Notwithstanding section 8.7.1 the manager may develop procedures for drilling within 300 mm of a bootleg or a hole containing explosives on the surface or within 150 mm underground.

(2) The manager shall submit these procedures to the inspector and to the OHSC at least 15 days in advance of implementing these methods and procedures.

Drilling in Loose Rock

8.7.3 No person shall drill in a loose rock produced by blasting unless

(1) the rock has been thoroughly examined to ensure that it does not have any holes containing explosives, or

(2) an engineered offset pattern is utilized to prevent overlaying of holes, and